

Sırrı Kaan Oğuzkan

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EDUCATION

- Bilkent University** Ankara, Turkey
• *B.Sc. in Computer Engineering* Sep 2023 – Present

EXPERIENCE

- Alkan Lab for Bioinformatics and Computational Genomics** Ankara, Turkey
• *Undergraduate Volunteer* February 2026 – Present
 - **CuCLARK GPU & CUDA Modernization:** Forked and modernized CuCLARK (GPU metagenomic classifier, originally CUDA 7.5 / single-arch); upgraded to **CUDA 11.8** with multi-arch support spanning **10** GPU generations (Kepler **sm_35** through Hopper **sm_90**, including K80, V100, A100, RTX 30xx/40xx, H100).
 - **Build System & Containerization:** Replaced legacy `install.sh` with a **CMake 3.20** build (C++17, CUDA 14, optional OpenMP) and authored a multi-arch `Dockerfile` published as `alkanlab/cuclark`; added a unified CLI with **6** subcommands wrapping the original scripts.
 - **Batch Classification Pipeline:** Diagnosed a byte-offset batching bug causing duplicate read assignments on small inputs and built a list-mode pipeline with resume, crash-recovery blacklist, and progress monitoring to classify **3000+** FASTA files in a single DB load.
- Look & Cash** Ankara, Turkey
• *Part Time Software Engineer* February 2025 – October 2025
 - **Ticket-Based Support System:** Built a ticket workflow for web/mobile (priority, comments, attachments, statuses). Shipped **13** REST endpoints, **4** ticket states, and **5** admin actions; reduced average admin handling time by **25%** via bulk actions and better filtering.
 - **Real-Time Mobile Support Chat:** Implemented real-time ticket chat on mobile with backend persistence and state sync. Achieved **300 ms** p95 message delivery in staging and **99.5%** delivery success under **150** concurrent chat sessions during load tests.
 - **Reporting & Graph Generation:** Built KPI reporting APIs over MongoDB (tickets, campaigns, scans, notifications) with date-range and campaign filters. Generated **14** dashboard datasets powering **4** chart types; cut report generation time from **6.2 s** to **1.9 s** via indexing.

PROJECTS

- Artifactum (CS319 – Software Engineering Project):** Owned artifact ingestion and LLM features for a MERN application.
 - **Gemini Integration:** Integrated Gemini through a service layer for artifact generation/analysis. Supported **6** prompt workflows; reduced LLM-call failures to **1%** using timeouts, retries, and structured parsing.
 - **Uploads & ZIP Parsing:** Built S3 multi-file uploads and ZIP bulk upload parsing with validation and safe partial failures. Processed ZIPs up to **50 MB** / **200** files; cut ingestion time from **40 s** to **15 s** using streaming + concurrency (**6** workers).
- NutriApp – Meal Recommendation Platform:** Built backend services for a meal-planning app under the Amazon University Engagement Program.
 - **REST APIs:** Implemented serverless REST APIs on AWS Lambda for core user flows. Built **16** endpoints; sustained in testing with p95 latency **450 ms**.
 - **Recognition:** Won the **People's Choice Award**. Top **1/7** teams.

SKILLS

- **Programming Languages:** C++, Java, JavaScript, TypeScript
- **Backend & Databases:** Node.js, Express, AWS Lambda, DynamoDB, MongoDB
- **Systems & DevOps:** C++, CUDA, CMake, Docker, Linux, Bash, Git
- **Core Topics:** Full-stack web development, REST API design, Authentication (JWT, RBAC), NoSQL data modeling, File uploads, Input validation, Logging & monitoring